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CXI EXPERIMENTAL STATION AT SOFTIMAX BEAMLINE MAX IV

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Abstract

A new Coherent X-ray Imaging (CXI) endstation has been designed and built at SoftiMAX, a soft X-ray spectro-microscopy beamline at MAX IV Laboratory. It features an in-vacuum rotatable detector stage with motorized movement over a 0-500 mm sample-detector distance range. This allows continuous positioning over scattering angles exceeding 120°. For longer distances up to 2 meters, fixed detector positions are supported via detachable vacuum extensions connected to angled ports on the vacuum chamber. This configuration enables a wide range of techniques, including X-ray Photon Correlation Spectroscopy (XPCS), scattering, diffraction and reflection. The internal mechanics are mounted on a granite base, mechanically decoupled from the vacuum chamber to ensure stability. A floor mounted rail system allows the entire endstation to be retracted, enabling quick installation of alternative setups. Final commissioning is expected in autumn 2025.

INTRODUCTION

The CXI experimental station is installed on the second branch of the SoftiMAX, a soft X-ray spectro-microscopy beamline at MAX IV Laboratory, the Swedish national synchrotron facility located in Lund. SoftiMAX operates at the 3 GeV storage ring, which provides exceptionally high average coherent flux due to the ring's low emittance. The CXI branch line uses a Kirkpatrick-Baez (KB) mirror pair to focus the beam to approximately 20 µm at the sample position. The KB mirror mechanics, based on an in-house design developed at MAX IV, are located approximately 2 m upstream from the sample and deflect the beam by 2 degrees both horizontally and vertically.

This endstation is dedicated to X-ray Photon Correlation Spectroscopy (XPCS), as well as diffraction and scattering experiments, and is designed with a very flexible detector geometry.

The experimental setup is housed in a large vacuum chamber designed for good accessibility, accommodating both the sample stages and a custom-built, in-vacuum detector stage. The detector can be continuously positioned within a 0-500 mm range from the sample while rotating across scattering angles exceeding 120°. For larger distances, up to 2 m, the detector can be mounted at discrete positions using detachable vacuum extensions (flight tube) connected to angled ports on the chamber walls.

Figure 1 shows a 3D model view of the CXI endstation assembly with a flight tube connected to one of the chambers angled ports. The design also includes a custom-built trolley to avoid floor collisions with the rail guides.

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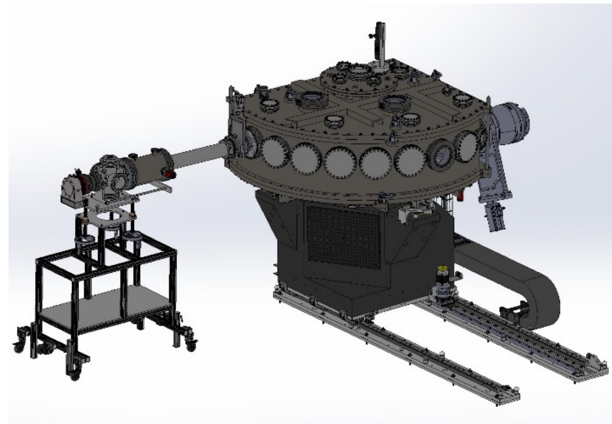

Figure 1: 3D view of the CXI endstation with example of a detachable flight tube on a trolley connected to one of the vacuum chambers angled ports.

The sample positioning system is divided into two subsystems: a coarse alignment stage based on stepper motors offering micron-level resolution and a fine positioning stage using piezo-driven actuators for nanometer-level precision. All internal mechanics are mounted on a granite base that is mechanically decoupled from the vacuum chamber to minimize vibration and drift. Additionally, the entire endstation can be retracted on floor-mounted rails to allow fast installation of alternative equipment or endstations brought by external users.

The following sections describe the design considerations, implementations and performance of each subsystem comprising the CXI endstation: The sample stages, the custom detector stage, the vacuum chamber and the mechanical support.

SAMPLE STAGES

The sample stages of the CXI endstation are designed to provide precise and accurate sample positioning. The sample stages consist of two subsystems: one with submicron resolution using stepper motor stages and a second subsystem with nanometer-level resolution using piezo driven motor stages (see Fig. 2). The sub-micron resolution stack is intended for rough alignment and includes x, y, z and rotation in y degrees of freedom. These stages are motorized using stepper motors and can provide a generous load capacity, ensuring stability and vacuum compatibility.

The nanometer resolution stack is intended for precision positioning and includes x, y, z and rotation in x and z axes degrees of freedom. Piezo motors are used to achieve nanometer resolution. These stages also include appropriate vacuum compatibility. The system includes an adequate general cable management system, with special attention to the cables from the nano-positioning stages, which are accommodated through the stepper motor Y rotation stage.

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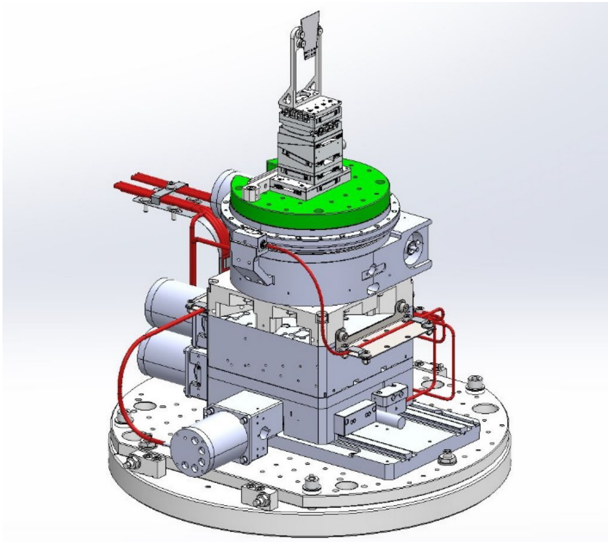

Figure 2: Stepper motor stages on the bottom alignment plate and piezo sample stages on the green top plate

Resolutions and ranges for the stages are described in Table 1. The stepper motor and detector stages include absolute encoders. The achieved specifications meet or exceed the design specification for the project.

Table 1: Stages

Stepper motor sample stages	Specifications
Degrees of freedom	X, Y, Z, RY
Resolution	10 µm, 1.725 µm, 10 µm, 0.087 mrad
Ranges	±15, ±5, ±50 mm, 360°
Piezo motor sample stages	
Degrees of freedom	X, Y, Z, RX, RY
Resolution	1 nm, <1 µ° (0.017 µrad)
Min. distance to sample	60.5 mm
Detector custom stages	
Degrees of freedom	X, RY
Resolution	10µm, 0.2mrad
Ranges	±250mm, 120°

DETECTOR CUSTOM STAGES

The custom detector stages in the CXI play a crucial role in providing the flexibility required for various experiments. The rotation around the y-axis, with a range of 120 degrees, allows the detector to be positioned at multiple angles relative to the sample. This rotation stage consists of curved ball rails driven by a low-backlash pinion and cam system, powered by an in-vacuum stepper motor.

Mounted on top of the rotation stage is a linear stage with a 500 mm travel range which houses the detector and auxiliary equipment such as the beam stop. This arrangement enables positioning of the detector as close as 60.5 mm from the sample. Figure 3 shows the stages on the detector rotation base plate.

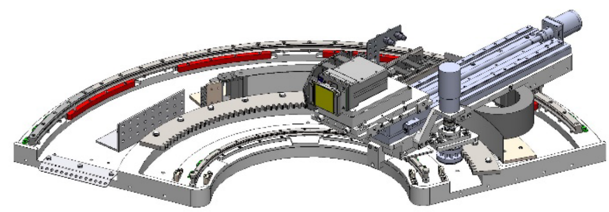

Figure 3: Rotation and linear stages for the detector custom stages.

Both stages are designed for precision (stated in Table 1) and repeatability, offering micron-level resolution, and are fully vacuum compatible to ensure reliable operation inside the vacuum chamber. The stages incorporate an effective cable management system with a rotative cable chain to accommodate the necessary signal and power cables and water-cooling hoses for the detector.

To ensure mechanical stability the base support of both stages are connected via rigid columns to the granite support block.

VACUUM CHAMBER

The vacuum chamber accommodates both the sample and detector stages, ensuring a stable vacuum environment with an operating pressure of 10^{-6} mbar or better. The pumping system consists of a 5-axis magnetically levitated turbopump, offering a pumping speed of up to 2150 l/s for N_2. Figure 3 shows view of the vacuum chamber with angled ports.

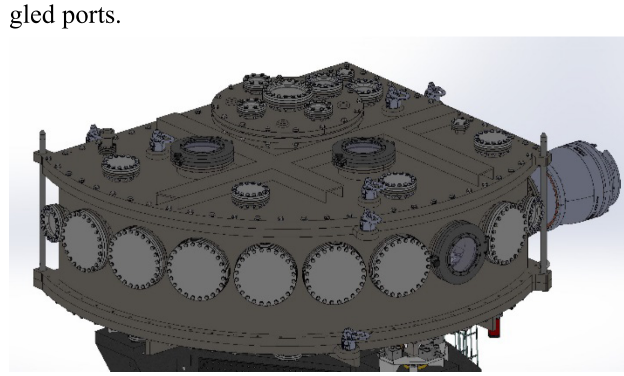

Figure 4: Vacuum chamber, angled ports and pump.

The chamber is equipped with a generously sized door that provides direct access to the sample environment (see Fig. 5) and enables maintenance access to the detector and custom detector stages.

It features a "sarcophagus" design, with a bottom plate, walls and top lid constructed as independent pieces each with its own lifting points (can be seen in Figs. 4 and 5). This modular design simplifies assembly and enhances flexibility during maintenance procedures.

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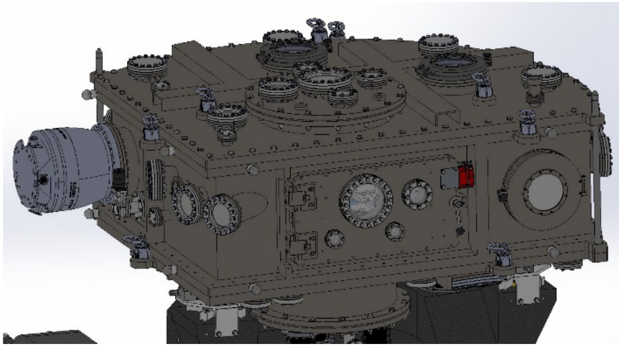

Figure 5: Vacuum chamber with rectangular port for direct access to the inside of the chamber.

To maximize mechanical stability, the vacuum chamber is supported independently from the stage supports. This decoupling is achieved using a system of bellows and rigid columns (see Fig. 6). The chamber is also equipped with all necessary vacuum ports, including those for pumping, venting, vacuum gauges, electrical feedthroughs and several spare ports for future use.

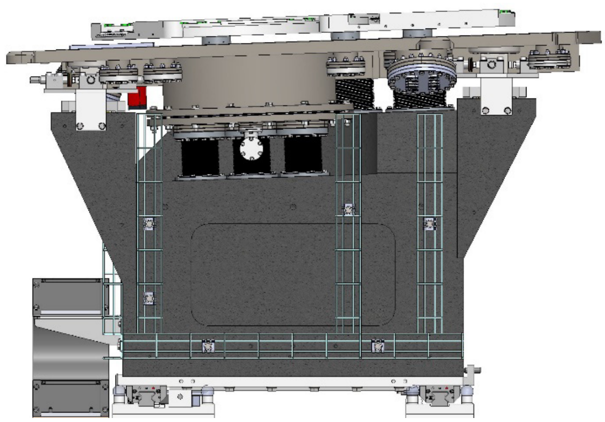

Figure 6: Granite block base decoupled from vacuum chamber and detector rotation base plate. Note the angle of the chamber bottom plate at 2°.

In addition, the chamber is designed to accommodate a customized flight tube extension (see Fig. 1) that houses a dedicated detector. The custom-built trolley floor collisions with the rail guides and is adjustable to always keep parallel to the scattering plane that is angled at 2°.

This detector can be positioned at various angles relative to the sample's y-axis through angled vacuum ports, allowing sample-detector distances of up to 2 m. A large vacuum port located above the sample position enables the integration of auxiliary equipment when needed.

MECHANICAL SUPPORT

The design of the support structure maximizes the mechanical stability of the system using granite as the preferred material due to its high rigidity and thermal stability. The support keeps the vacuum chamber independently supported from the sample and detector stages ensuring that vibrations or mechanical drifts due to pumping in the vacuum chamber are not transferred to the detector stage (see Fig. 6).

Additionally, the whole system can move out of the beam position, maintain at least one meter of space around the sample position. This allows external user-based endstations to be installed. For example, XPCS measurements, Fourier Transform Holography of magnetic thin films, or the development of a TIMEPIX imaging detector for the soft X-ray regime, which have already been used. This is achieved thorough a floor rail system (see Fig. 1) driven by a rack and pinion mechanism powered by a stepper motor. The movement is designed to be repeatable to minimize the setup time when returning the station to its original position. The support structure also includes all necessary components for effective cable management (see Fig. 6), such as cable trays and a large cable chain.

To guarantee safety during the motion of the entire endstation, measures have been implemented to ensure that movement is performed only under operator control. A dead man switch has been installed to ensure that motion occurs only when there is continuous input from the operator inside the hutch helping to prevent accidental or unintended movements.

CONCLUSIONS

The mechanical design of the CXI endstation meets the required specifications in respect to precision, stability, vacuum, safety and flexibility. The stages inside the main vacuum chamber give a wide range of motion both in rotation and distance from the sample. With the added flight tube and custom trolley, a detector can be placed at discrete angles up to 2 m from the sample. The floor rails where the granite block and chamber are mounted give the beam line a possibility to accept external user set-ups.

We look forward to the commissioning during the autumn of 2025.